

Normal and Hermitian.

Lemma. Let V be an inn. prod space and T is a Hermitian operator.

(If exists) Every eigenvalue of T is real.

Moreover, eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue C of T corresponding to α ,

$$C \cdot \langle \alpha, \alpha \rangle = \langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \langle \alpha, C \cdot \alpha \rangle = \bar{C} \cdot \langle \alpha, \alpha \rangle$$

Since α is non-zero, $C = \bar{C}$, or C is a real number.

And, take α_1, α_2 be eigenvectors corresponding to C_1, C_2 , with $C_1 \neq C_2$.

Then,

$$C_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle C_1 \alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, C_2 \alpha_2 \rangle = \bar{C}_2 \cdot \langle \alpha_1, \alpha_2 \rangle$$

Since C_2 must be real, $\bar{C}_2 = C_2$. Now, since $C_1 \neq C_2$,

$$(C_1 - C_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0 \text{ implies } \langle \alpha_1, \alpha_2 \rangle = 0.$$

Thm. In finite-dimensional inn. prod space,

Every Hermitian operator has an eigenvector.

Proof. If the given field is $\mathbb{C}$, then the characteristic polynomial must have a root in $\mathbb{C}$ by the Fundamental Theorem of Algebra. So it has an eigenvector.

Now, assume that the given field is the $\mathbb{R}$.

Let T be a Hermitian on V over $\mathbb{R}$, and β be an orthonormal basis for V .

Let $A = [T]_{\beta}$. Since $T = T^* = T^t$, $A = A^* = A^t$. Define an inner product on $\mathbb{C}^n$ as:

$$\langle X, Y \rangle \stackrel{\text{def}}{=} Y^* X$$

Then, a linear operator $L_A: \mathbb{C}^n \rightarrow \mathbb{C}^n: X \mapsto AX$ is Hermitian, because

$$\langle L_A(X), Y \rangle = \langle X, L_A^*(Y) \rangle \text{ and } \langle AX, Y \rangle = Y^* AX = \langle X, A^* Y \rangle = \langle X, AY \rangle.$$

By the FTA, L_A has an eigenvalue, and that must be real, by the above Lemma.

Now, $L_A(X) = AX = CX$ for some real number C and a non-zero vector X in $\mathbb{C}^n$.

or $\det(CI - A) = 0$, $A - CI$ is not invertible, and $A - CI$ has only real-entries,

so $(A - CI)X = 0$ has a $\underset{\text{non-zero}}{\text{real solution}} X$.

Lemma. Let V be a fin. dim inn. prod space, and T be a linear operator on V .

If $W \subset V$ is T -invariant, then $W^\perp$ is T^* -invariant.

Proof. Given $\beta \in W^\perp$, and $\alpha \in W$, $\langle T^*(\beta), \alpha \rangle = \langle \beta, T(\alpha) \rangle = 0$, by $T(\alpha) \in W$. $\square$

Diagonalizable Criterion of linear operator on inner product space.

Thm. Let T be a linear operator on fin. dim inner prod space V .

If T is Hermitian, then there is an orthonormal basis for V consisting by eigenvectors of T .

Proof. Let T be a Hermitian. Using induction for $\dim V = n$.

If $n=1$, trivial. And, for n , using the facts that:

- 1). T has an eigenvector α_1 .
- 2). $\langle \alpha_1 \rangle \oplus \langle \alpha_1 \rangle^\perp = V$.
- 3). $\langle \alpha_1 \rangle^\perp$ is T -invariant, so $\langle \alpha_1 \rangle^\perp$ is T^* -invariant, and $T = T^*$.

Corollary. In $\mathbb{R}$ -Inner Product space,

T is symmetric if and only if there is orthonormal basis for V consisting by eigenvectors of T .

Lemma. Let T be a normal operator on fin. dim inner prod space V . Given $\alpha \in V$,

α is eigenvector for T corresponding to $c \in \mathbb{F}$ iff α is eigenvector for T^* corresponding to $\bar{c} \in \mathbb{F}$.

Proof. If L is normal, then

$$\|L(\alpha)\|^2 = \langle L(\alpha), L(\alpha) \rangle = \langle \alpha, L^*L(\alpha) \rangle = \langle \alpha, LL^*(\alpha) \rangle = \langle L^*(\alpha), L^*(\alpha) \rangle = \|L^*(\alpha)\|^2.$$

And, any scalar $c \in \mathbb{F}$, $T - cI$ is normal, by

$$(T - cI)(T - cI)^* = T^*T - \bar{c}T - cT^* + c\bar{c} = T^*T - \bar{c}T - cT^* + c\bar{c} = (T - cI)^*(T - cI).$$

So, $\|(T - cI)(\alpha)\| = \|(T - cI)^*(\alpha)\|$ gives the result.

Thm. Let T be a linear operator on fin. dim V s.t. V , and β be an orthonormal basis for V .

Suppose that $A = [T]_\beta$ is upper-triangular. Then,

T is normal if and only if A is diagonal.

Proof. Let T is normal, and $\beta = \{\alpha_1, \dots, \alpha_n\}$. Since A is upper triangular,

$$T(\alpha_1) = A_{11}\alpha_1 \text{ and so } T^*(\alpha_1) = \bar{A}_{11}\alpha_1 = \sum_{i=1}^n (A^*)_{i1} \cdot \alpha_i = \sum \bar{A}_{1i} \cdot \alpha_i$$

So, $\bar{A}_{12} = \bar{A}_{13} = \dots = \bar{A}_{1n} = 0$. Similarly, all non-diagonal entries are zero.

The Converse is trivial. ~~QED~~

Now, since every linear operator on $\mathbb{C}$ -vector space has a basis s.t. the representation is triangular. Using Gram-Schmidt process to this, we obtain that every normal in $\mathbb{C}$ is unitary diagonalizable.

Summary.

$T: V \rightarrow V$, V fin. dim inn. prod over F

T is Normal and $F = \mathbb{C}$

In $\mathbb{C}$, T is Hermitian $\Rightarrow T$ is unitary equivalent to a diagonal matrix $\Downarrow$

In $\mathbb{R}$, T is symmetric $\Leftrightarrow T$ is orthogonal equivalent to a diagonal matrix

In $\mathbb{C}$, Hermitian $\Rightarrow$ Normal $\Leftarrow$
 $\Downarrow$
Symmetric